

Appendix Z

Detailed Thematic Analysis of Opportunities to Improve LitTraceAI

Z.0.1 Enhancing Usability and User Interaction

This theme reflects expert suggestions aimed at improving both the functionality of LitTraceAI and the quality of interaction with generated content. However, the suggested improvements operated at different levels. Some recommendations focused on increasing trust and verification capabilities, others aimed to support idea generation and development of the experts' own work, while additional suggestions related to accessibility, personalisation, and flexibility in how the generated content could be produced and explored.

Some experts highlighted the importance of features supporting trust and verification of generated content. 4 out of 22 experts requested the possibility of highlighting the original source sentences directly within the uploaded papers. One expert stated:

Also, in the provenance section, it [LitTraceAI] should take me to the paper and highlight the text where the source was extracted from, similar to how Zotero does it. (Expert 7)

Another expert explained:

I would have liked for the original sentences in the provenance, the tool [LitTraceAI] to apply it to the original document that I gave you. I mean, I, as a human being, I am very visual! Well, I am very visual and on top of that, distrustful. I prefer to check the information in the original document. (Expert 20)

Similarly, 2 out of 22 experts requested direct access to source PDFs through citations, with one expert suggesting that *“in the citations you can have access to the PDFs to double check that the AI [LitTraceAI] does not make them up”* (Expert 18). This request appeared to be related to verifying that the citations generated by LitTraceAI corresponded to real source papers rather than fabricated references. Although the tool already provides the original source sentences used to generate the text, some experts still considered this insufficient for verification purposes. These responses suggest that, for certain experts, trust in the generated content depended on being able to inspect the extracted sentences within the original paper and directly access the source PDFs to verify that the citations were genuine.

Other suggestions focused on helping experts obtain a rapid overview of the generated literature and use it as a starting point for developing their own research ideas. Visual representations of the literature were highlighted by 6 out of 22 experts. Experts described the need for diagrams, conceptual maps, and structured overviews that could provide a general picture of the generated content, particularly when dealing with complex topics. One expert explained:

A graphic layout that shows how the ideas are connected, or at least the constructs, so that the researcher has an overall view of how the process began and how the new methodologies being applied are branching out. Something like a mind map! (Expert 7)

Another expert noted that *“because the topic is complex and comprehensive, a visual representation will help me to have a general picture”* (Expert 10). Similarly, Expert 24

requested that LitTraceAI should “*produce an outline of the important aspects ... like a mind map or something like that.*” These responses suggest that some experts valued visual synthesis features because they could support quicker digestion of the generated information and serve as a foundation for developing their own literature review.

Additional functionality supporting idea exploration and workflow integration was also identified, particularly regarding identification of research gaps for idea exploration, and citation formatting and export of generated outputs for workflow integration.

2 out of 22 experts requested flexible citation formatting, with one expert recommending that LitTraceAI should “*include different citation formats*” (Expert 6). Since the tool currently provides citations in IEEE format, this response may indicate that certain experts, particularly from disciplines such as the social sciences, valued having the option to switch to other citation styles such as APA.

Another 2 out of 22 experts requested filtering options for sentence classifications, particularly for identifying research gaps. One expert explained: “*I would like to filter all the research gaps to have them as a foundation in my work*” (Expert 15). This response may indicate that some experts valued filtering features because they could help them more quickly identify research opportunities within a topic.

Similarly, experts also highlighted the importance of easily exporting the generated content. Drawing on a previous negative experience with Elicit, Expert 22 explained that exporting results, particularly tables, had been “*a nightmare*”, and suggested that LitTraceAI should allow users to export the generated content “*in a click*” into formats such as PDF or Word documents. In relation to the tool, the same expert also requested that “*you can download the results*” and preferred them “*to export to Endnote*” (Expert 22). These responses may indicate that some experts valued export functionality that would allow them to quickly download and reuse the generated outputs within their existing research workflows.

Additional suggestions focused on extending interaction with the generated outputs. Expert 5 proposed incorporating audio functionality so that “*it [LitTraceAI] reads to you across*

all sections”, and also requested the option to change the language. In a follow-up conversation, the expert explained that *“certain users who are more auditory”* may adopt applications more easily when audio features are included. The expert also noted that, as a non-native English speaker, *“listening to the content gives me an idea of the correctness of the text because I can hear if the written text make sense or not.”* Regarding language customisation, the expert explained that *“for some research areas like in law or sociology the research is based on Latin America so the information is written in Spanish.”* Separately, Expert 21 suggested displaying authors and publication years alongside provenance information, stating that *“the papers from which the provenance is based should display the authors and years of the papers”* (Expert 21). Although each of these suggestions was raised by 1 out of 22 experts, they illustrate additional ways in which experts expected LitTraceAI to better support interaction with the generated content.

Finally, some experts requested features that would provide greater flexibility in how the literature review is generated. One expert suggested allowing LitTraceAI to process images, stating: *“Let the tool feed itself with images”* (Expert 12). This request was related to allowing the tool to incorporate visual diagnostic results for Respiratory Sarcopenia as a complement to the generated content. However, the images were described as complementary information and did not replace the expert’s expectations regarding the textual content generated by LitTraceAI.

Another expert proposed enabling users to provide their own headings before generation:

One simple addition to provide the headings yourself and with the set of papers, the tool should generate the related work section. (Expert 16)

This response may indicate that some experts valued having greater control over the structure of the generated content, particularly when they already had predefined topics or conceptual maps guiding the literature review. However, as discussed in Section ??, some experts also appeared to value automatic heading generation because it could provide additional ideas to explore. This finding may require further research to identify in which

situations experts prefer predefined headings and in which situations they prefer LitTraceAI to automatically generate them.

Overall, most of the suggested improvements appeared to focus on extending the functionality, usability, and flexibility of LitTraceAI rather than addressing deficiencies in the quality of its generated content. The main exception concerned features related to trust and verification, such as source highlighting and access to source PDFs. As discussed in Section W.0.1, trust appears to be a particularly relevant factor in the automatic generation of literature reviews, since certain experts needed mechanisms to verify that the generated content and citations accurately reflected the original sources.

Z.0.2 Enhancing Transparency and Academic Integrity

This theme captures expert suggestions aimed at improving the transparency, traceability, and verifiability of LitTraceAI’s generated content. Experts highlighted limitations in how the tool communicates evidential support, transforms source material, and ensures academic integrity.

Concerns regarding evidential support thresholds were mentioned by 4 out of 8 experts. Experts appeared to expect stronger criteria for determining when generated sentences could be considered fully-supported by the literature. Expert 13 emphasised that *“the enormous potential that the tool already has is right there in the provenance”*, but questioned statements classified as fully supported when *“only supported by one paper.”* The expert further suggested that stronger support classifications could require broader corroboration across studies because *“a single paper can be an isolated case.”* Similarly, Expert 17 argued that *“the generated statement should be supported by more than one sentence whenever applicable”*, suggesting that some experts may prefer support classifications grounded in multiple supporting citations. Expert 8 also stated that *“the tool should cite whenever necessary”* because *“sometimes, a citation is not robust enough to support a claim”*, further recommending that LitTraceAI should explicitly indicate when *“the sentence is missing citations”*.

Expert 26 further explained:

*There are cases where citations are required, general ideas appear such as: **there is extensive evidence that identification with the organisation is related to pro-organisational unethical behaviour.** Sure, that is true! But I cannot leave it like that in a literature review. I have to indicate what that evidence actually is and cite the studies that demonstrate it. In other words, explain that multiple studies have found that the greater the organisational identification, the higher the probability that the person will adopt pro-organisational unethical behaviours, and accompany that statement with the corresponding references.*

This observation differs from the previous concerns regarding insufficient support from one citation or sentence. Instead, it highlights situations where generated sentences include broad claims about evidence without providing any supporting citations. Together, these observations may indicate that some experts expected LitTraceAI not only to strengthen evidential corroboration, but also to explicitly identify when supporting citations are missing.

In a follow-up conversation, Expert 21 reflected further on how LitTraceAI handles partial evidential support and secondary referencing. Regarding partial support classifications, the expert stated that the tool “*detects important gaps and makes comparisons and contrasts, but sometimes it makes strong claims*” even when statements are classified as Partial Support.

Expert 21 explained further that “*this assertion should not be so categorical because it is not fully supported by the source*” and suggested that LitTraceAI should “*soften the claim a little*” by using more cautious expressions such as “*to some extent...*” This observation may indicate that some experts expected the linguistic strength of generated sentences to align more closely with their level of evidential support, particularly when the provenance classification is Partial Support.

Expert 21 also reflected on the effect of secondary referencing on citation precision:

Let's say it is a fairly strong assumption because what the tool [LitTraceAI] is doing is paraphrasing what another author said, but not the original author. Thus, I do not know to what extent that is acceptable and it should be differentiated. The tool should indicate when it is quoting what the original author said and when it is giving me a secondary citation. That is, in this case, which is a secondary citation, that should be one of the criteria that lowers the precision of the citation.

This observation may indicate that some experts expected the criteria for citation classification to account not only for evidential support, but also for whether citations originated from primary or secondary sources. Together, these reflections from Expert 21 suggest reconsidering how citation classifications are determined and communicated to users.

Plagiarism detection was raised by 2 out of 8 experts. Experts expected LitTraceAI to identify plagiarism not only in generated text but also within source materials. Expert 1 suggested that the tool should detect “*plagiarism from the authors of the original papers*”, while Expert 11 referred to detecting “*whether the content it generates is plagiarism.*” However, plagiarism detection is not an inherent limitation of LitTraceAI itself, as the tool was designed around verifiability rather than factuality. Furthermore, the implemented text transformation mechanisms may already support manual plagiarism detection to some extent by allowing users to inspect how the generated text was transformed from the original sources.

One expert out of 8 requested a higher level overview of provenance accuracy and source content usage. Expert 25 suggested that “*the AI [LitTraceAI] should display a general graph reflecting the percentage of information retrieved from the set of papers, as well as the overall accuracy of the provenance.*” This request aligns with the concept of citation precision proposed by Liu et al. (2023b), defined as the proportion of generated citations from a given text that support their associated statements. While LitTraceAI already provides citation classification at the sentence level, it did not include overall citation precision metrics or the percentage of sentences used from the set of papers in the generated content due to the

limited time available for experts to evaluate the tool. Nevertheless, the current architecture would allow these features to be incorporated immediately and evaluated in future studies. Providing an overall view of citation precision and source sentence usage may also foster user trust by enabling experts to assess the verifiability of the overall generated content more efficiently.

Only 1 out of 8 experts raised concerns regarding text transformation. Expert 27 stated:

If you want to turn this into a draft of the literature review, you will really have to engage with what kinds of text transformations are necessary. The generated text does little text transformation from the original papers.

This observation suggests a particular concern from Expert 27 regarding the level of paraphrasing and synthesis performed by LitTraceAI. However, as no other experts raised similar concerns, this does not appear to represent a broader limitation identified across the expert sample.

Z.0.3 Enhancing Analytical Capability, Citation Accuracy, and Literature Coverage

This theme captures expert suggestions aimed at improving the analytical functionality, citation clarity, and completeness of the AI-generated literature review. While experts recognised the usefulness of LitTraceAI for generating literature review, they also identified additional features that could strengthen its usefulness. These suggestions included expanding the set of reviewed papers through automated identification of related studies, supporting the generation of hypotheses based on the reviewed literature, and improving the differentiation between original and indirect citations to strengthen clarity in attribution.

A recurring suggestion concerned the automatic expansion of the literature base (3 out of 5 experts). In contrast to the previous observation about the value of a selected corpus. Experts proposed that LitTraceAI should identify additional relevant studies beyond those

initially provided by the researcher. Expert 4 stated that the tool should *“recommend papers that are important and that I did not include in the list I sent you.”* Likewise, Expert 6 suggested that the tool should:

Go to scientific databases and connect these papers with others that are in the databases. That is, show me a list of related papers.

Similarly, Expert 11 explained that *“The tool should have access to search engines and recommend similar and complementary studies to be included.”* These responses may indicate that experts valued features that could assist them in identifying additional relevant literature related to the source papers.

One expert (1 out of 5 experts) also suggested extending the analytical functionality of LitTraceAI through automatic hypothesis generation. Expert 6 stated that the tool should *“deduce the hypothesis and write it”*, adding that *“given this literature review, the hypothesis is this.”* This response may indicate an expectation that LitTraceAI could support analytical tasks in addition to synthesising literature.

Another expert (1 out of 5 experts) highlighted the importance of distinguishing direct references from indirect citations for maintaining academic rigour. Expert 21 explained:

It should differentiate citations from the text and citations of citations. This is important for the academic rigour in academic writing. The text should always refer to the original author.

This observation may indicate that experts viewed citation provenance as an important element of academic rigour in AI-generated writing. The concern that *“the text should always refer to the original author”* suggests that indirect citations were perceived as potentially weakening transparency regarding the origin of evidence, which may affect researchers’ ability to verify claims and evaluate the reliability of the generated content.

Z.0.4 Enhancing Literature Representation

This theme refers to expert-identified improvements aimed at strengthening the tool’s capacity to support deeper interpretation and synthesis of the literature. While mentioned by a single expert, this perspective is noteworthy. The expert has research expertise in knowledge graphs and related areas that intersect with literature review generation, making the observation particularly relevant to the design of AI-supported literature review tools. The expert described the need for representations that move beyond surface-level summaries and make relationships between claims, concepts, and studies more explicit.

Expert 27 explained:

What I mean is that you would extract claims, identify where each claim comes from, and map the logical structure... An argument map or discourse map of this kind is what I actually want, because it exposes the underlying knowledge.

The expert’s statement indicates that current AI-generated literature review sections may not provide sufficient support for higher-level reasoning about the literature. Rather than focusing only on coherent prose generation, the expert emphasised the importance of representing the conceptual structure underlying the reviewed studies. The expert further argued that “*having a graph representation of the reasoning structures in the literature could support many different writing tasks*”, suggesting that structured representations may support interpretation and additional literature-related writing activities. The expert also noted that “*once the structure is represented, the usual problems around paraphrasing disappear*”, although provenance “*is still important and must be addressed.*” This observation indicates that the expert associated deeper knowledge representations with broader possibilities for analysing and working with the literature.